

CREDIT CARD FRAUD DETECTION

PROJECT WORKFLOW

From transaction data to fraud-risk prediction

MILESTONE 02 • WORKFLOW

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“Data → Learn → Score → Flag”



01 The End-to-End Workflow

A single pipeline transforms raw transaction records into an actionable fraud-risk decision.



02 Workflow A — Build the Learning Pipeline

The model only becomes useful after the transaction data is made consistent, balanced and measurable.

DATA PREPARATION

01 **INPUT** Transaction records + fraud label

02 **CLEAN** Missing / duplicate / invalid records

03 **TRANSFORM** Scale numeric features; prepare inputs

04 **BALANCE** Class weights or controlled resampling

05 **SPLIT** Train / validation / test sets

FEATURE SPACE

time • amount • transaction features • encoded variables

Output → a consistent numerical representation for ML models

MODEL TRAINING

LOGISTIC REGRESSION

baseline / interpretable

DECISION TREE

rule-based patterns

RANDOM FOREST

robust ensemble

OPTIONAL: XGBOOST

strong benchmark

03 Workflow B — Evaluate, Select & Predict

Model selection is driven by fraud-detection quality, not accuracy alone.

PRECISION

How many flagged cases are actually fraud?

RECALL

How much fraud did we catch?

F1

Balance between precision + recall

ROC-AUC

How well does the model rank risk?

MODEL COMPARISON

A

TEST PREDICTIONS

Generate probabilities + classes

B

COMPARE METRICS

Focus on recall, precision & F1

C

SELECT WINNER

Best practical trade-off

D

DEPLOY

Save trained model

NEW TRANSACTION

→ FRAUD SCORE

→ RISK LEVEL

→ FLAG / ALLOW / REVIEW

Example: probability 0.91 → HIGH RISK → send for verification

04 Final System View

This is the workflow we will implement as the semester project prototype.



Prototype outcome: a trained fraud classifier + prediction interface + evaluation report